Canadian Community Health Survey (CCHS)

Household weights documentation

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1. Introduction

This document describes the weighting process used in the creation of the household weight for the Canadian Community Health Survey (CCHS). The process is very similar to the process for other data products but here the product represents households rather than individuals. In using this weight, users should note that the survey is designed to collect information from individuals and not households. Certain steps are taken in the design to ensure that the sample is representative of different demographic groups and this may affect how well the sample represents households of different compositions. Also, since the calibration for the household weight is done at the provincial level, estimates are considered reliable only at the national and provincial level.

This weight should only be used for household-level variables. Such variables are those where the person-level responses clearly represent the household.

2. Weighting Overview

To ensure estimates produced from survey data are representative of the target population and not just the sample itself, users must incorporate the survey weights in their calculations. A survey weight is given to each household included in the final sample, that is, the sample of households where a person has answered the survey. Each weight corresponds to the number of households in the entire population that are represented by the household of the respondent.

The CCHS has recourse to three sampling frames for its sample selection: an area frame acting as the primary frame and two frames made up of telephone numbers used to complement the area frame. Since only minor differences differentiate the two telephone frames in terms of weighting, they are treated together and referred to as being part of the telephone frame.

Depending on the need, one or two frames are used for the selection of the sample within a given health region (HR). When two frames are used, the weighting strategy treats both the area and telephone frames independently to come up with separate household-level weights for each of the frame used. These weights are then combined into a single set of weights through a step called *integration*. After some adjustments, this integrated weight becomes the final household weight.

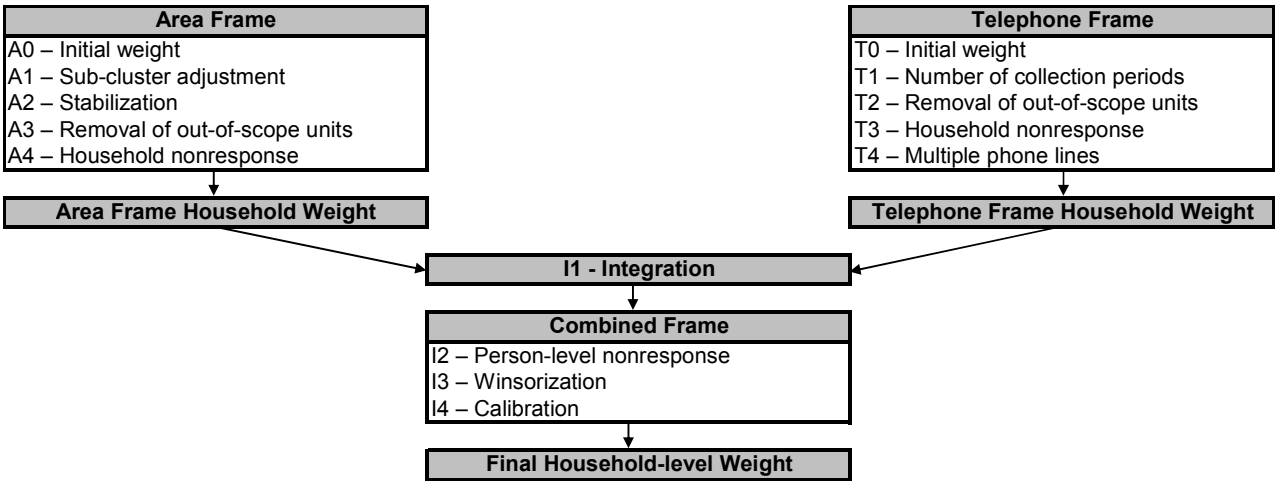
3. Sample weighting

As mentioned earlier, units from both the area and telephone frames are treated separately up to the integration step. The household weighting steps up to and including the integration of the frames are the same as the steps from the CCHS person-level weight. Please refer to the CCHS User Guide for more information about these steps. The final three weighting steps deviate from the person-level strategy and are explained in sub-sections 3.1-3.3.

Although the two frames are used to cover the three territories, the sampling methods used are slightly different from those used in the provinces. These modifications affect the weighting of these three regions, and they are reported in sub-section 3.4.

Diagram A presents an overview of the different adjustments that are part of the weighting strategy. A numbering system is used to identify each adjustment and will be used throughout the section. Letters *A* and *T* are used as prefixes to refer to adjustments applied to the units on the *A*rea and *T*elephone frames respectively, while prefix *I* identifies adjustments applied from the *I*ntegration step onwards.

Diagram A: Household weighting strategy overview



3.1 Person-level nonresponse (I2)

In some cases, interviewers obtain a household roster and select a member of the household but do not conduct an interview either because they cannot get in touch with the individual, or because that selected person refuses to be interviewed. Such individuals are defined as person nonrespondents, and an adjustment factor must be applied to the weights of person respondents to account for this nonresponse. In the case of the household weight, this person-level adjustment must still be applied since the selected person has not responded for the household.

Using a methodology similar to the one used in the treatment of household nonresponse (see the CCHS User Guide, Section 8.2 - A4), an adjustment is applied within response homogeneity groups. In this process, a scoring method is used to define a response probability based on characteristics available for both respondents and nonrespondents. Since it is the selected individual that did not respond for the household, the characteristics of this person are used in determining the response probability for the household.

The probabilities are grouped into response homogeneity groups within each province and the following adjustment factor is calculated within each group:

$$\frac{\text{Sum of weight I1 for all selected persons}}{\text{Sum of weight I1 for all selected persons who respond}}$$

Weight I1 is multiplied by the above adjustment factor to produce weight I2. Nonrespondents are dropped from the weighting process from this point onward.

3.2 Winsorization (I3)

Following the series of adjustments applied to the respondents, some units may come out with extreme weights compared to other units of the same domain of interest. These units could have a large impact on the variance. In order to prevent this, the weight of these outlier units is adjusted downward using a "winsorization" trimming approach.

3.3 Calibration (I4)

The last step in obtaining the final CCHS household weight is calibration. Calibration is done to ensure that the sum of the final weights corresponds to known household counts based on the Census. Monthly household counts are available and are defined at the province by household size level. The household sizes used are 1 person, 2 persons, and 3 or more persons.

Calibration is performed using the program CALMAR. This program adjusts the weights to ensure that the sum of the final household weights corresponds to the average of the monthly household counts at the province by household size level. At the same time, since data is collected throughout the year, the weights are adjusted to ensure that time periods are equally represented within the sample. For example, the households collected in one 2-month collection period represent $1/6^{th}$ of the overall sum of weights.

The weight I3 is adjusted to obtain the final weight I4. Weight I4 corresponds to the ***final CCHS household weight*** that can be found on the household weight file with the variable name WTS_MHH.

3.4 Particular aspects of the weighting in the three territories

For the Territories, differences in the weight adjustment steps up to and including integration are summarized in the CCHS User Guide. After integration, the only difference concerns the level at which calibration is performed. The period control is set to be two 6-month periods. An additional control for the distribution of households inside / outside the capital cities is used.

3.5 Creation of the Share weight (WTS_SHH)

Along with the master file and PUMF, which contain all CCHS responding persons, a share file is created which contains only a portion (>90%) of the original CCHS responding persons. The individuals on this share file have agreed to share their data with certain partners. To compensate

for the loss of some respondents from the file, the weights of these "sharers" must be adjusted by the factor:

$$\frac{\text{Sum of weight I2 for all respondents}}{\text{Sum of weight I2 for all respondents agreeing to share their data}}$$

Similar to the nonresponse adjustments, this factor is calculated within homogeneity groups, where in this case, households with similar estimated propensity to share will be grouped together. Weight I2 for sharers is multiplied by the above adjustment factor to produce a share weight. Winsorization and calibration, similar to adjustments I3 and I4, are applied to the share units. The final weight after these adjustments is called WTS_SHH.

3.6 The Food Security Module

The Food Security Module (FSC) is one of the few CCHS modules where it is appropriate to use the household weights. For 2011-2012, the FSC module is part of common content and is asked in all provinces and territories. For the 2009-2010 collection period, the FSC module was part of optional content and was not selected for some of the provinces. Given this change, comparing estimates between 2011-2012 and 2009-2010 may not be appropriate at the national level. For more information on optional content selection, please refer to the appendices of the User Guide.